Name: _____

WORKSHEET

Degrees and radians

1 Convert the following angles into radians.

a 180° _____

b 45° _____

c 120° _____

d 60° _____

e 30° _____

f 225° _____

g 90° _____

h 315° _____

2 Convert the following radian measurements into degrees.

a $\frac{\pi}{2}$ _____

b $\frac{\pi}{3}$ _____

c $\frac{3\pi}{4}$ _____

d $\frac{5\pi}{6}$ _____

e π _____

f $\frac{5\pi}{12}$ _____

g $\frac{11\pi}{6}$ _____

h $\frac{13\pi}{9}$ _____

3 Convert the following radian measurements into degrees and minutes.

a 0.5678 _____

b 2.7906 _____

c 4.6635 _____

d 1.0872 _____

e 5.8713 _____

f 3.1027 _____

4 Convert the following angles into radians correct to four decimal places.

a 32°45′ _____

b 172°16′ _____

c 349°23′ _____

d 291°55′ _____

e 93°09′ _____

f 201°37′ _____

5 Draw a unit circle and mark in the position of:

a π

b 90°

c $\frac{3\pi}{2}$

d 45°

e $\frac{\pi}{3}$

f 2π

g 30°

h $\frac{5\pi}{4}$

i 315°

j 150°

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6 Write down the size of the angles in the correct measurement as shown on this unit circle. Write a, c, e, h and i in degrees and the others in terms of π.

$a =$

$b =$

$c =$

$d =$

$e =$

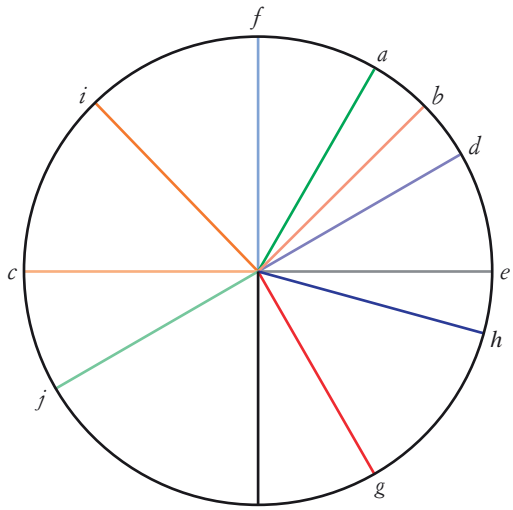
$f =$

$g =$

$h =$

$i =$

$j =$



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Answers

1 **a** π

b $\frac{\pi}{4}$

c $\frac{2\pi}{3}$

d $\frac{\pi}{3}$

e $\frac{\pi}{6}$

f $\frac{5\pi}{4}$

g $\frac{\pi}{2}$

h $\frac{7\pi}{4}$

2 **a** $90°$

b $60°$

c $135°$

d $150°$

e $180°$

f $75°$

g $330°$

h $260°$

3 **a** $32°32'$

b $159°53'$

c $267°12'$

d $62°18'$

e $336°24'$

f $177°46'$

4 **a** 0.5716

b 3.0066

c 6.0979

d 5.0949

e 1.6258

f 3.5189

5 **a** $60°$

b $\frac{\pi}{4}$

c $180°$

d $\frac{\pi}{6}$

e $360°$

f $\frac{\pi}{2}$

g $\frac{5\pi}{3}$

h $330°$

i $135°$

j $\frac{7\pi}{6}$

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